Cross-Modal Face Retrieval: Matching Hand-Drawn Sketches to Photographs via Metric Learning

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Abstract

This paper presents a deep learning approach to the cross-model retrieval problem of matching hand-drawn facial sketches to corresponding photographs. Using the CUHK Face Sketch Database (CUFS), we implement a Siamese Network architecture based on a **ResNet-50** backbone. To address the significant domain gap and limited training data (**131** identities), we employ a "batch-hard" triplet loss strategy and a strategic layer-freezing approach. Here, embeddings of matching identities are pulled toward each other via batch-hard triplet loss, while dissimilar pairs are repelled, achieving a domain-invariant alignment. Our final model achieves a Rank-10 accuracy of **58.6%** on a 29-identity test set, significantly outperforming random chance **34.5%** and a contrastive-loss baseline. We identify input normalization and metric learning strategy as the two most critical factors in cross-modal feature alignment.

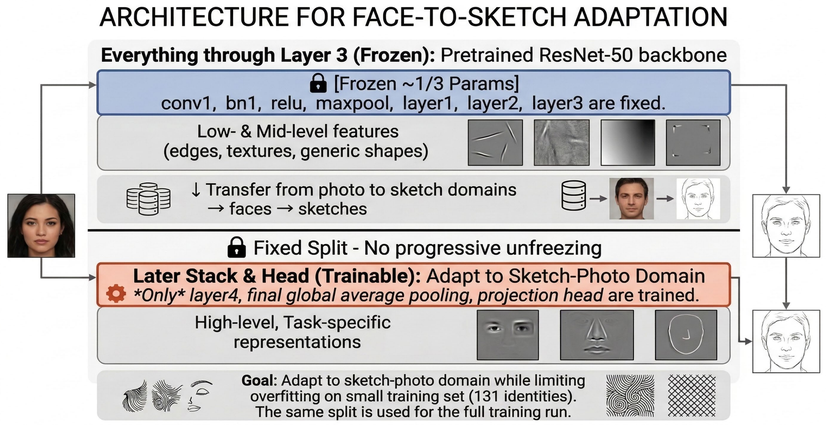
Methodology

Architecture

- **Feature Extractor:** After the final average pooling layer (layer 4's final residual block), we retain the 2048-dimensional feature vector.
- **Lightweight 2-Layer Projection Head:** Compresses these features into our target 128-dimensional embedding space: 2048 → 512(with BatchNorm and ReLU, Dropout 0.5) →128.
- **L2 Normalization:** Final embeddings are normalized. This constrains optimization to angular distances, stabilizing training by removing magnitude variance and enabling cosine similarity computation:

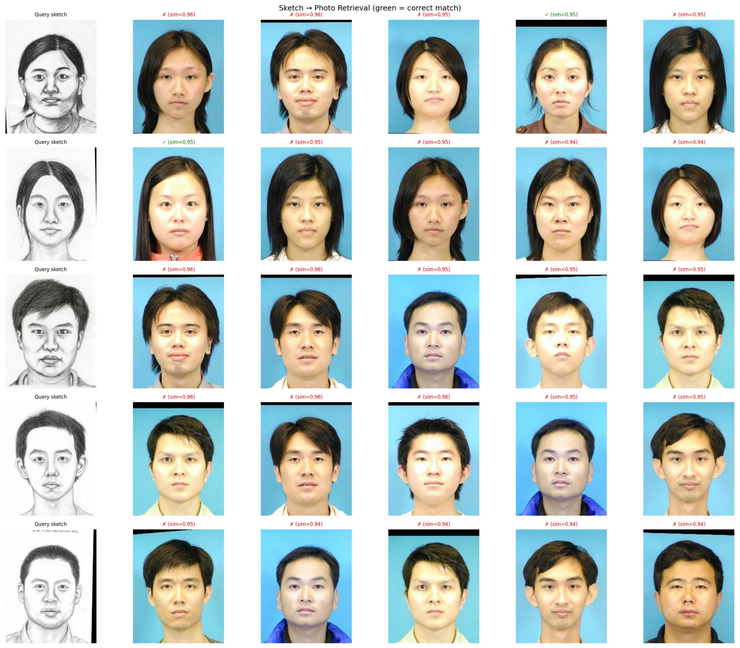
Layer Freezing Strategy

- **Latter Stack:** Fine-tuning only the latter stack (layer4) and the projection head are responsible for high-level, task-specific representations.



Result

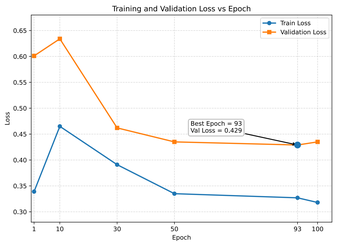
Retrieval Visualization



For each query sketch (left column), the top-5 most similar photographs are shown. Green = correct match, red = incorrect.

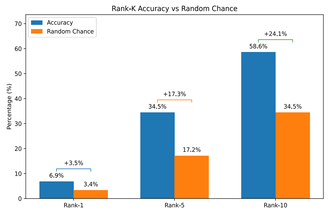
Training Process

Parameter	Value
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	5×10^{-3}
LR schedule	Cosine annealing $\rightarrow 10^{-6}$
Batch size	32
Epochs	100



Checkpointed at epoch 93 based on the lowest validation loss.

Evaluation



- **Rank-1** (6.9%, 2/29): Correct photo ranked #1 for 2 sketches. (2.0x improvement).
- **Rank-5** (34.5%, 10/29): True match appears in top 5 for 10 sketches.
- **Rank-10** (58.6%, 17/29): Best metric. (1.7xrandom baseline of 10/29 = 34.5%).
- The transition from Contrastive to Triplet Loss provided a **3.4%** boost while fixing the normalization provided a **17.3%** boost.

Version	Loss Function	Normalization	Rank-10 Accuracy
V1 (Baseline)	Contrastive	Simple	37.9%
V2 (Fixed Norm)	Contrastive	ImageNet	55.2%
V3 (Final)	**Triplet Loss**	**ImageNet**	**58.6%**

Table 2: Model evolution across versions (Rank-10 accuracy)

Conclusion

We have demonstrated that a **Siamese ResNet-50** architecture, optimized with batch-hard triplet loss, partial layer unfreezing, and aggressive augmentation, successfully bridges the sketch-photo domain gap on the challenging CUFS dataset. Despite training on only **131 identities**, our model achieves **Rank-10** accuracy of **58.6%**, 1.7x random baseline, systematically surfacing true matches in the top 10 candidates for 17/29 test sketches.

Key insights emerge from our ablation studies:

(1) Strategic layer freezing prevents overfitting while preserving low-level feature transferability;

(2) Batch-hard mining maximizes learning efficiency from sparse triplets;

(3) domain-invariant embeddings emerge even without explicit sketch-photo synthesis.

Challenge and Future Work

Challenges

Small dataset: 131 training identities with 1 sketch and 1 photo each is extremely limited for deep learning.

Domain gap: Sketches are grayscale line drawings while photos are full-color.

Future Work

Face-Specific Backbone: Replacing ResNet-50 with a model pretrained on ArcFace or VGGFace2 would likely provide a massive performance gain, as those models are alreadyoptimized for facial geometry.

Domain-Specific Batch Norm: Implementing separate Batch Normalization layers forthe sketch and photo branches could help the model better handle the different pixel-leveldistributions of the two domains.

References

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